



Fu-Jen Catholic University  
Department of Electrical Engineering



# Ch0 Computer System Overview

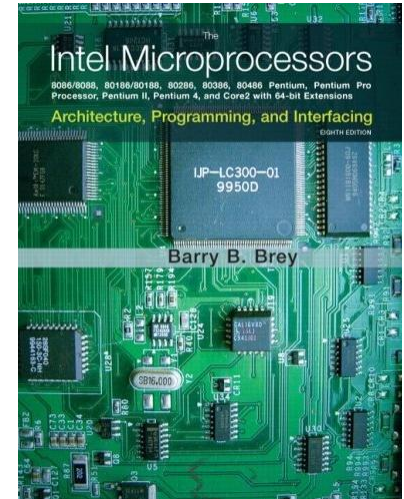
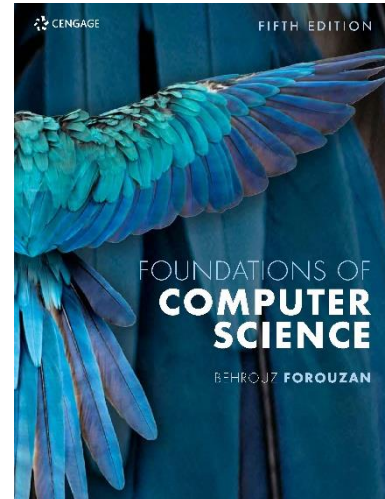
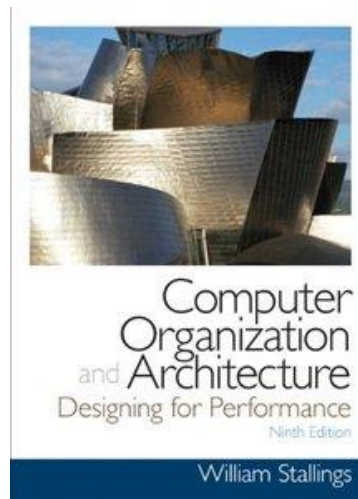


**Louis Chuang**  
**Dept. of EE, ENCL**  
**Fu Jen Catholic University**



# References

- William Stallings, “Computer Organization and Architecture, 9e,” Prentice Hall, Pearson Education (ISBN-13: 978-0-13-293633-0), 2013.
- Behrouz Forouzan, “Foundations of Computer Science, 5e,” Cengage Learning (ISBN: 978-1-4737-8732-2), 2023.
- Barry B. Brey, “The Intel Microprocessors : Architecture, Programming, and Interfacing, 8e,” Prentice Hall, Pearson Education (ISBN-13: 978-0-13-714094-7), 2009.





# Intellectual Property Rights

- 版權所有(Intellectual Property Rights) :
  - 輔仁大學電機工程系。
  - 僅供教學使用。
  - 不重製、不散布、不公開播放！
- Supplemented by: Yue-Ru Chuang (Louis).



# Computer Organization (1)

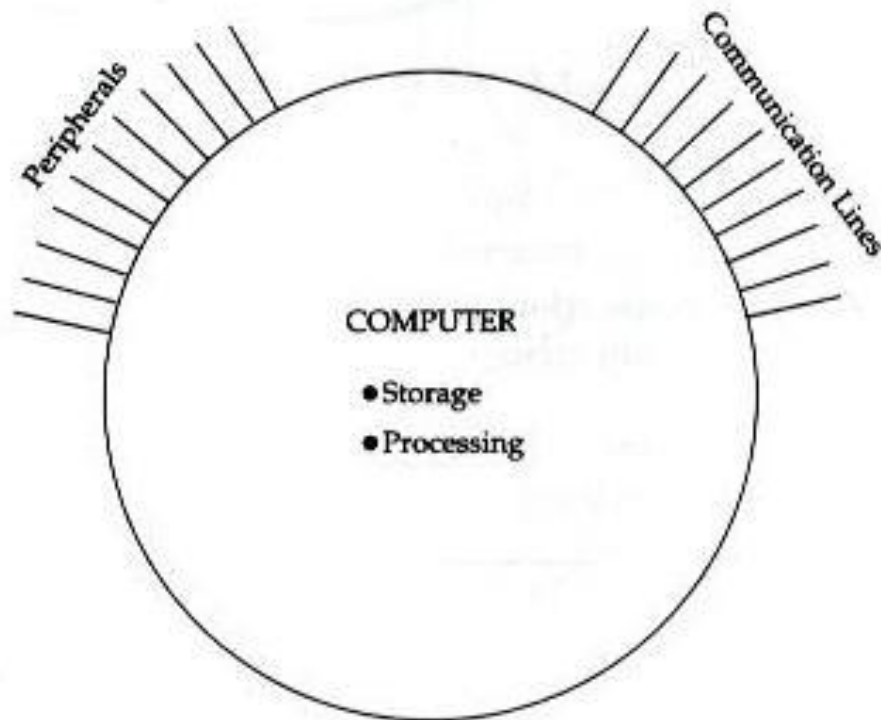
- ❖ 3 Parts:
  - ❖ (1) Computer (System).
  - ❖ (2) Peripheral devices.
  - ❖ (3) Communication lines.





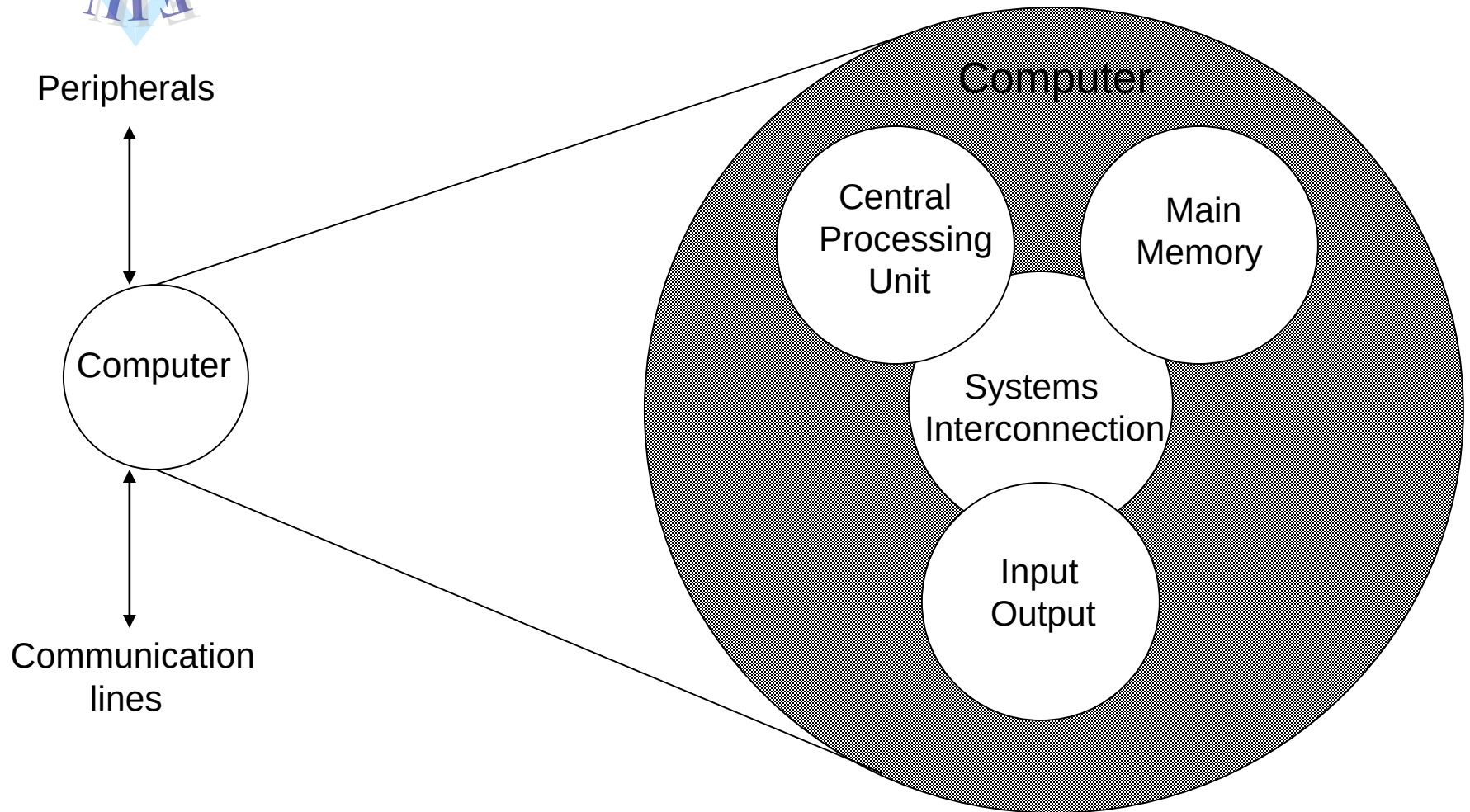
# Computer Organization (2)

- ❖ In general, all of its linkages to the external environment can be classified as (1) peripheral devices and (2) communication lines.





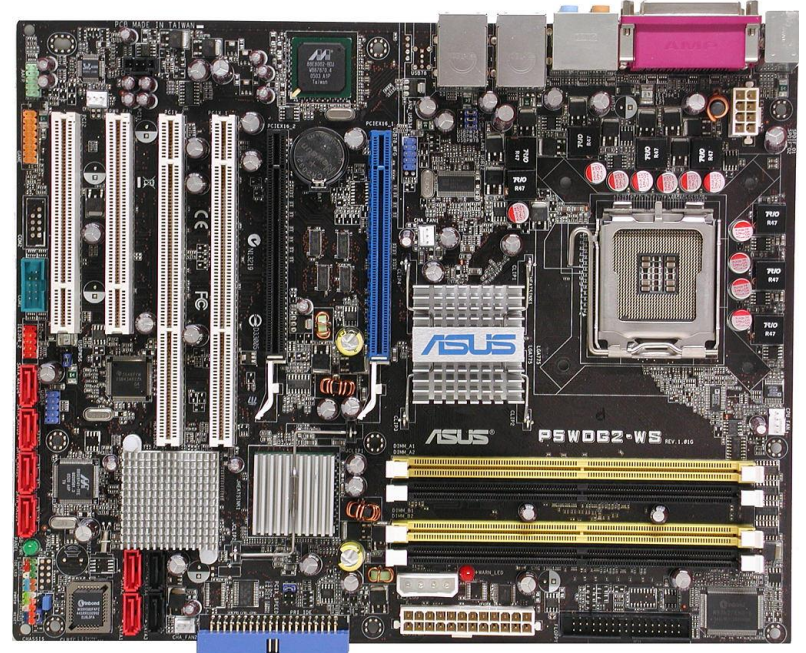
# Computer Organization (3)





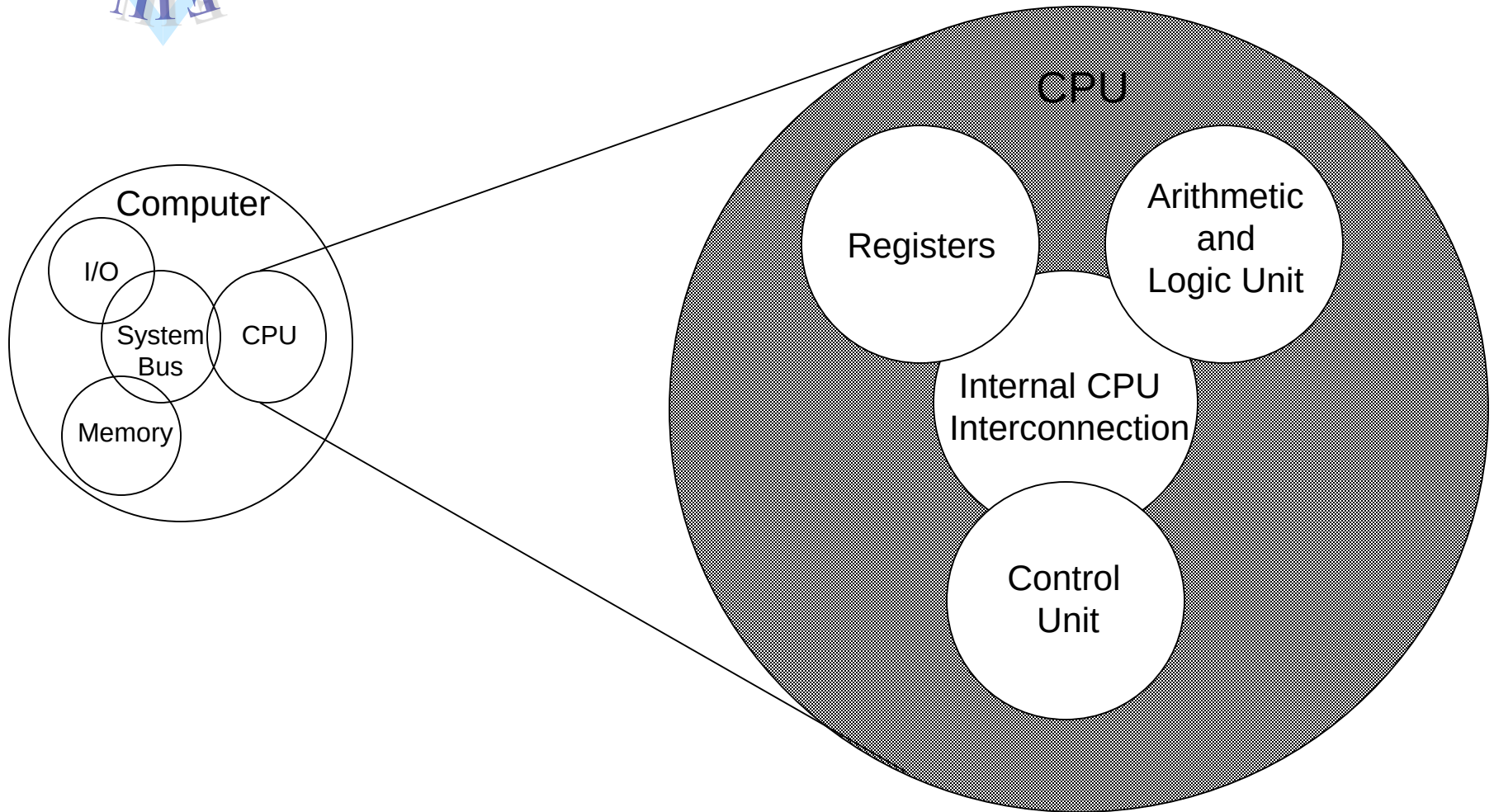
# Computer Organization (4)

- ❖ The principal elements of a computer system:
  - (1) Central processing unit (CPU).
  - (2) Main memory.
  - (3) I/O system.
  - (4) Interconnection mechanisms.





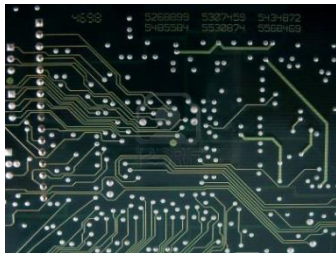
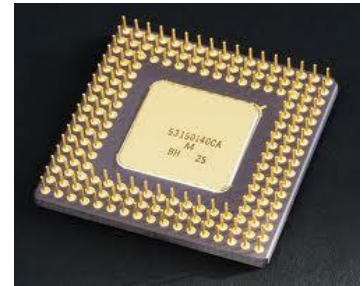
# CPU & Bus (1)





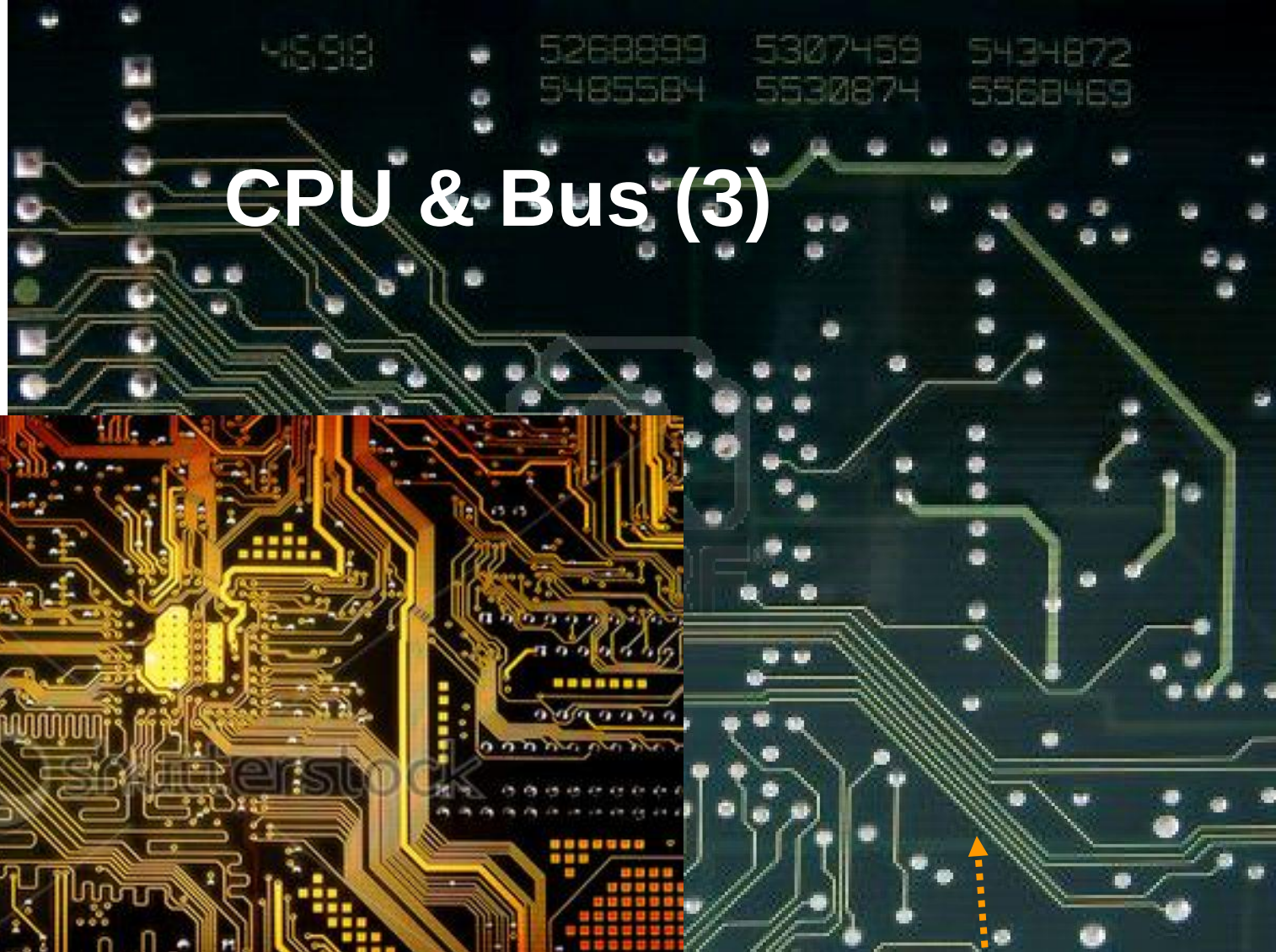
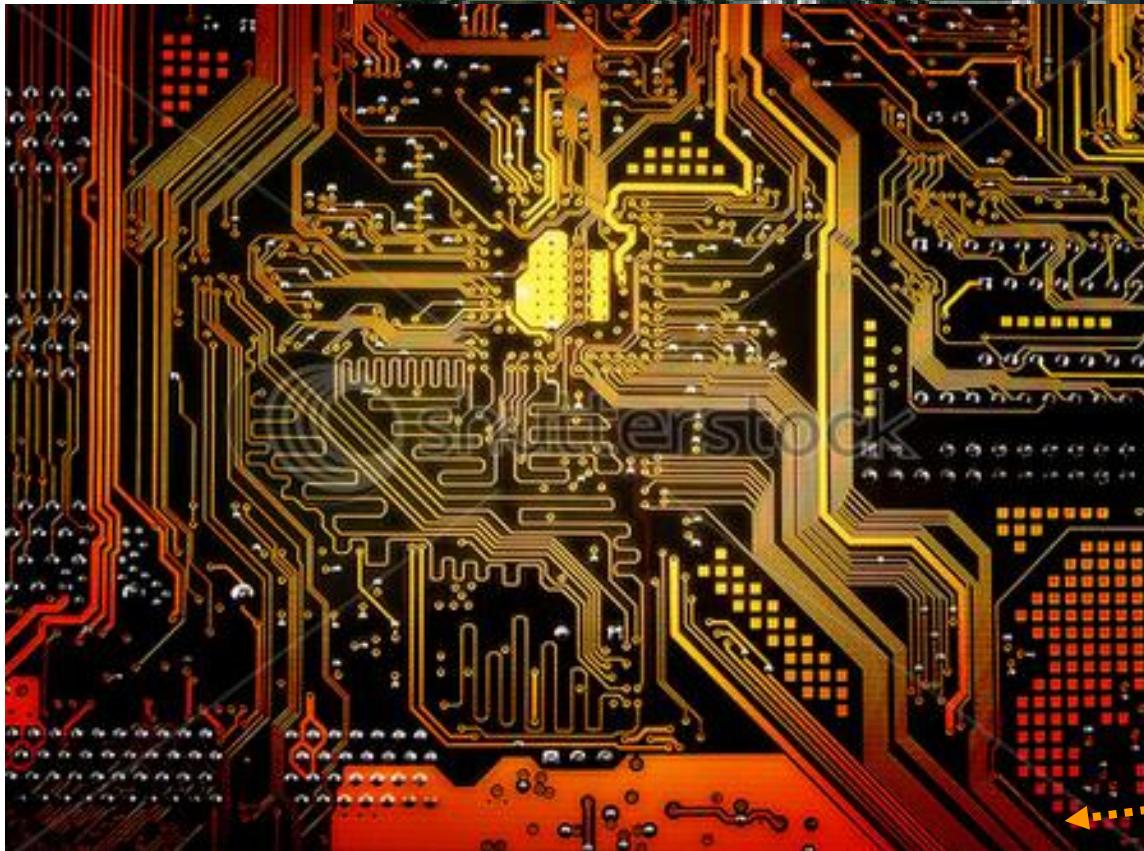
## CPU & Bus (2)

- ❖ The CPU consists of
  - (1) Control unit (CU).
  - (2) Arithmetic and Logic unit (ALU).
  - (3) Internal registers.
  - (4) Interconnection facilities.
- ❖ Three kinds of interface in CPU.
  - (1) Addressing bus.
  - (2) Data bus.
  - (3) Control bus.





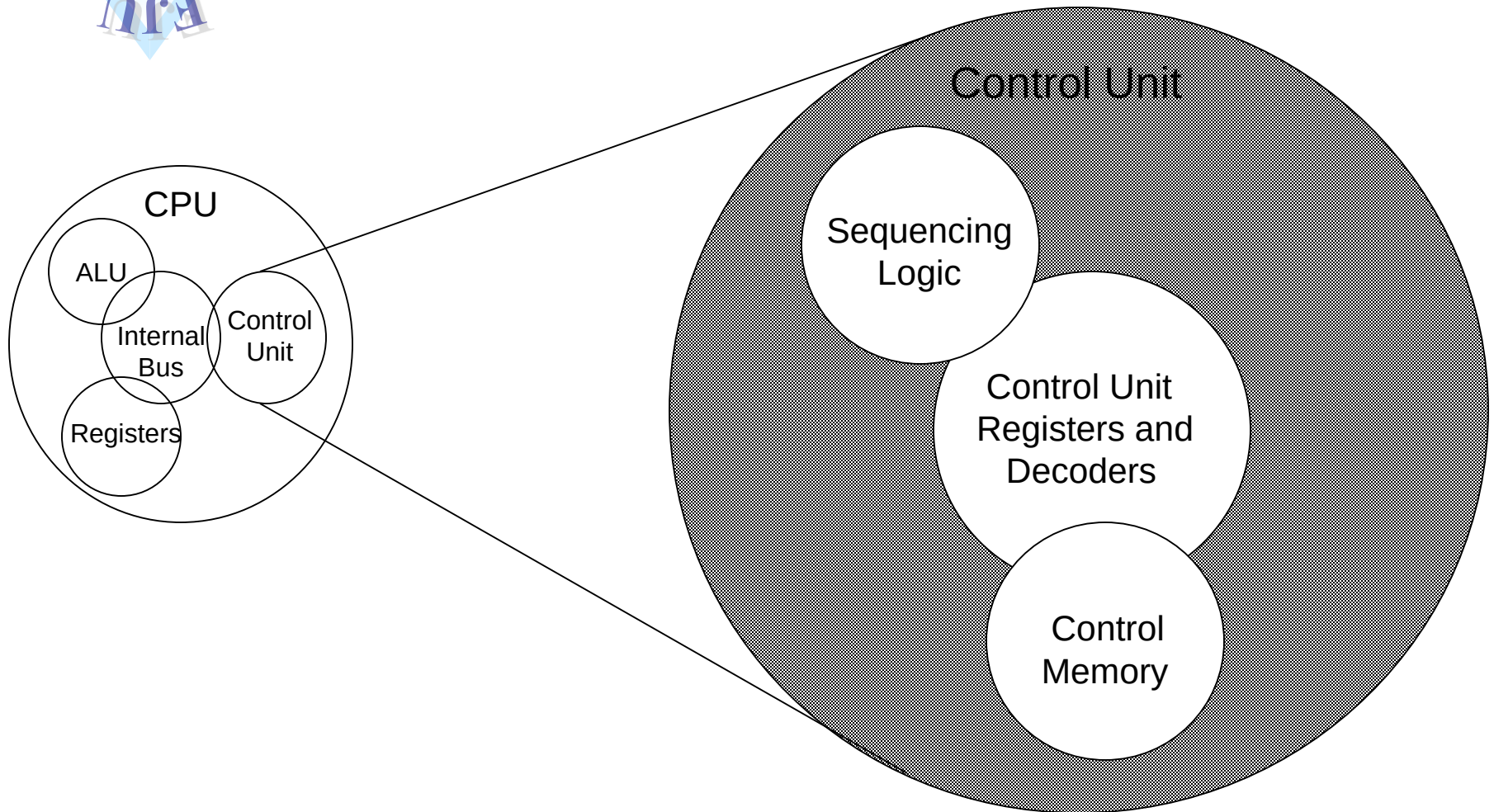
# CPU & Bus (3)



- ❖ Bus on PCB (Printed Circuit Board).

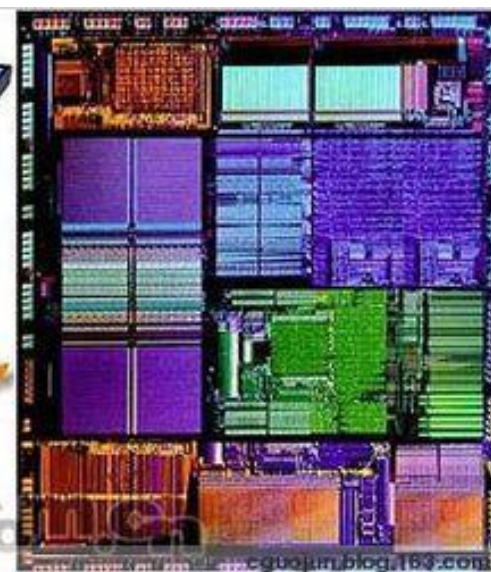
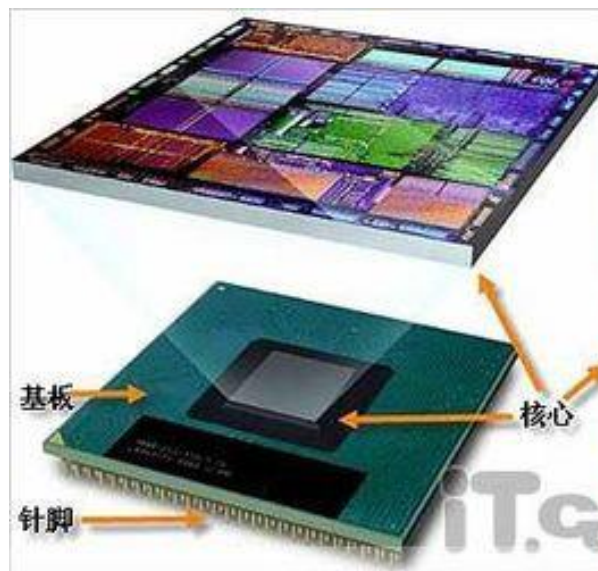
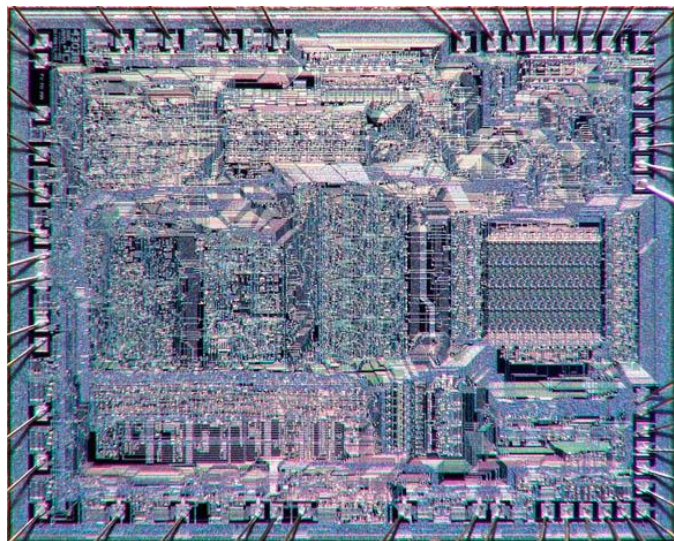
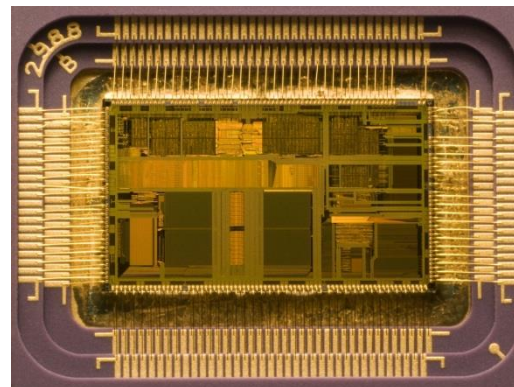
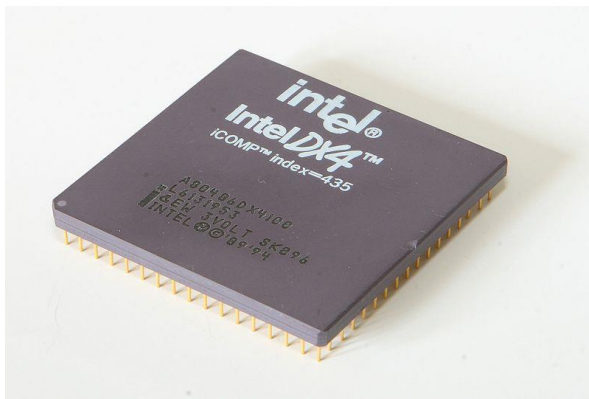


# CPU & Bus (4)





# CPU & Bus (5)

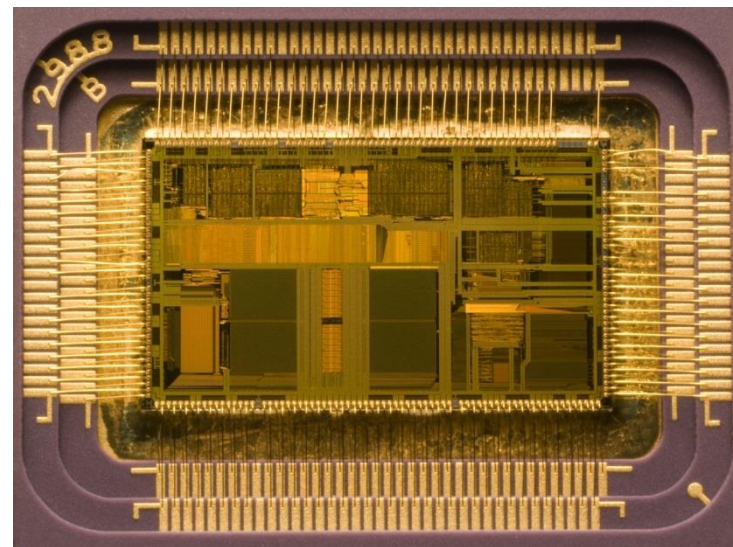
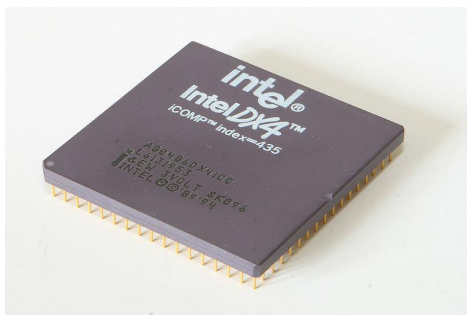
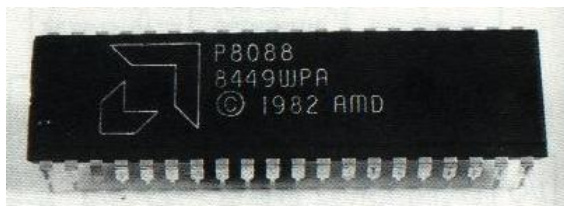




# 中央處理單元 (CPU)



- Intel CPU: 8088, 8086, 80286, 80386, 80486, Pentium ... (IBM).
- Motorola CPU: 6800, 68000, ... G3, G4 (Apple-Macintosh).
- AMD (Advanced Micro Devices) CPU.
- Other CPU (embedded system): RDC, Realtek, Marvell, Star, Atheros, ARM, ...
- CPU 設計趨向二~三層式的快取記憶體 (L1, L2, L3 Caches)，以彌補 CPU 與 Main memory 之間的速度差。
  - CPU速度快，Main memory速度較慢。





# 金價狂飆

## 晶片封裝廠有志一「銅」

- 在晶片封裝領域正掀起一片技術改革潮，可望大幅降低封裝成本與晶片價格，並催生更廉價的終端電子產品；但這波改革與炒得正熱的 3D 晶片——即所謂的矽穿孔(through-silicon-vias, TSV)晶片堆疊技術——關聯性不大，而是原本以黃金(gold)為主要材料的打線接合(wirebonding)技術，突然轉向採用價格較低的銅。
- 多年來，打線接合一直是晶片產業所採用的主要封裝技術，並以細黃金線來連結IC元件、製作接腳；但黃金價格近年來不斷飆漲，幅度甚至已達200%，讓很多(但不是全部)晶片廠商有了個好藉口轉換技術，紛紛改用銅線進行打線接合。據估計，採用銅線的封裝成本比採用黃金線便宜了至少三成。
- 包括：博通(Broadcom)、聯發科(MediaTek)、意法半導體(ST)與德州儀器(TI)等廠商，都悄悄提高了採用銅線接合的元件產量。而晶片封裝廠如日月光(ASE)、矽品精密(SPIL)與新科金朋(STATS ChipPAC)，也正大力推廣銅線接合製程技術。
- 電子工程師專輯2010/9/13  
([http://www.eettaiwan.com/ART\\_8800620263\\_480302\\_NT\\_1f4a4f19.HTM?click\\_from=8800059753,9223546984,2010-09-13,EETOL,ARTICLE\\_ALERT](http://www.eettaiwan.com/ART_8800620263_480302_NT_1f4a4f19.HTM?click_from=8800059753,9223546984,2010-09-13,EETOL,ARTICLE_ALERT)).



# CPU、GPU、NPU、TPU (1)

- CPU (Central Processing Unit)：中央處理器(單元)。
  - CPU中央處理器是讓計算機進行運作的中央指揮中心，也可稱為計算機的大腦。
- GPU (Graphic Processing Unit)：圖形處理器(單元)，通常為顯示卡上的處理器。
  - 又稱為電腦、平板及手機的第二顆心臟，執行繪圖運算工作的微處理器，執行分析深度學習和機器學習演算法尤其有用。
  - GPU顯示卡則是在中央指揮中心以下，負責繪圖運算工作的部門。因此CPU負責的是發送指令，而GPU則是接收指令，進行繪圖與顯示的幾何運算。
- 一文看懂CPU, GPU, NPU, TPU是什麼？  
(<https://blog.qnap.com/zh/%E4%B8%80%E6%96%87%E7%9C%8B%E6%87%82-cpu-gpu-npu-tpu-%E6%98%AF%E4%BB%80%E9%BA%BC/>).



# CPU、GPU、NPU、TPU (2)

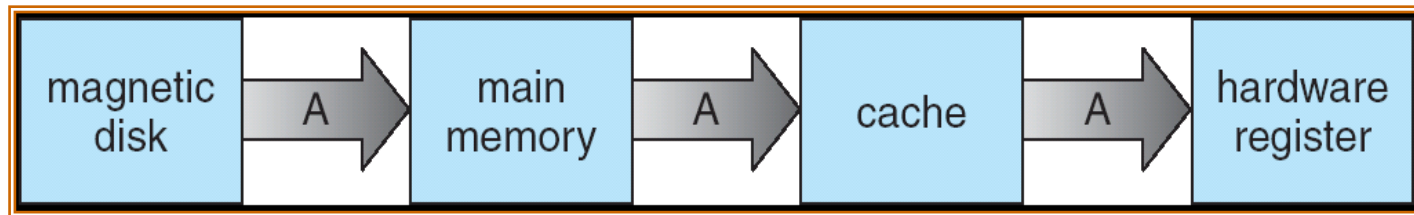
- NPU (Neural Network Processing Unit)：神經網路處理器(單元)。
  - NPU是專為加速AI運算應用而設計，具備節能的特性，可滿足長時間的使用，適合持續處理 AI運算任務。
- TPU (Tensor Processing Unit)：張量處理器(單元)。
  - TPU是Google開發專門用於加速機器學習的處理器。與GPU相比，TPU 被設計並運用於大量的低精度計算。Google於研究表明在使用神經網路推斷的 AI 運算任務中，TPU的效能是當代GPU和CPU的15到30倍，但是生產的廠商少，市場供不應求，因此價格昂貴。
- Intel® Core™ Ultra 處理器提供的三個專用引擎(CPU、GPU與NPU)皆可協助解鎖AI的威力、打造身歷其境的影像效果體驗，以及實現高效能低功耗的處理方式(2024. 9)。





# CPU vs. Cache vs. Memory vs. HD (1)

- CPU (電路)：速度最快。
  - Register (暫存器)：容量最小(16-bit/32-bit)，速度次快，成本最高。
  - Cache (快取記憶體)：容量次小(128KB/256KB/512KB/...)，速度慢一點，成本次高。
  - Memory (記憶體)：容量稍大(1GB/2GB/4GB/...)，速度更慢一些，成本稍低。
  - Hard-disk (HD, 硬碟)：容量大(300GB/500GB/...)，速度最慢，成本最低。
- Migration of integer A **from disk to register.**





# CPU vs. Cache vs. Memory vs. HD (2)

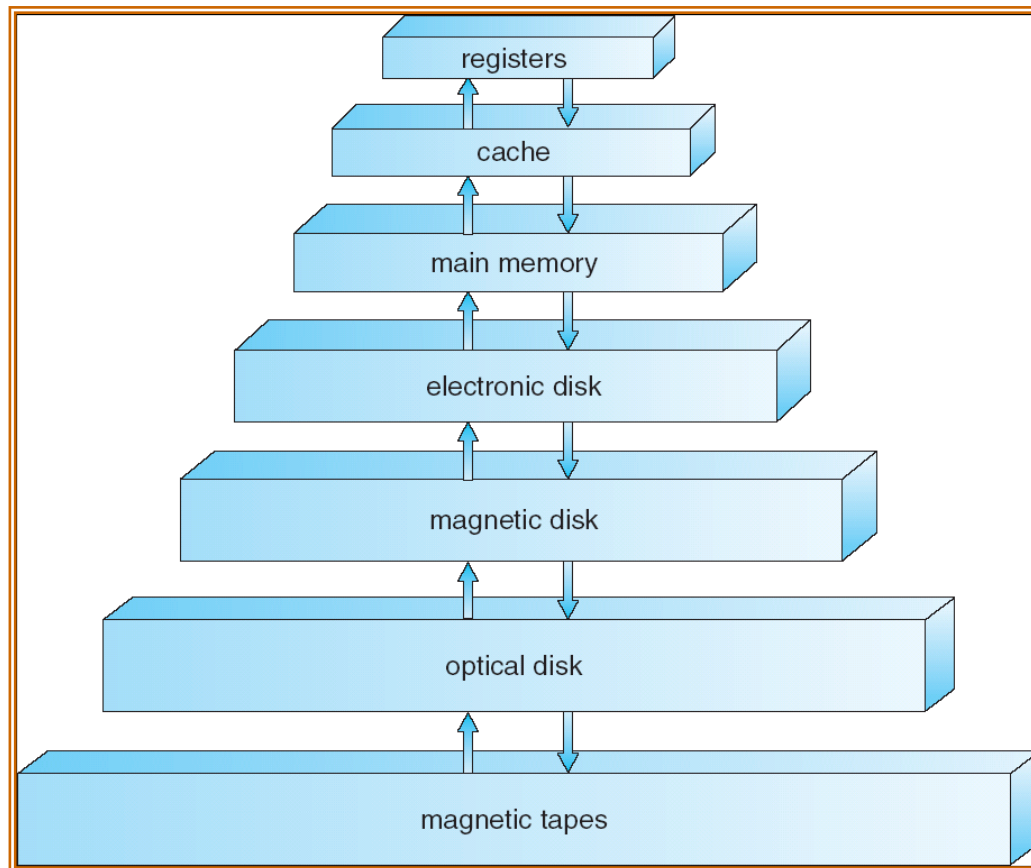
- **Spooling**：指的是一個資料傳輸過程中先將它放置在一個臨時工作區，讓另一個會使用到它的設備/程序可以在稍後的時間點再進行處理。而這個臨時工作區通常會是一個文件或存儲設備。
  - 原因：因為設備之間是以不同的速率來存取資料。
  - 例如：將記憶體當作是一個非常大的緩衝器，CPU不直接從硬碟上讀取資料(硬碟速度相對較慢)，而是先將硬碟中的輸入資料先存入記憶體中(記憶體速度相對較快)，然後CPU再從記憶體上讀取。

如此，可以彌補CPU高速但是硬碟傳送資料速率卻很緩慢的缺點。



# CPU vs. Cache vs. Memory vs. HD (3)

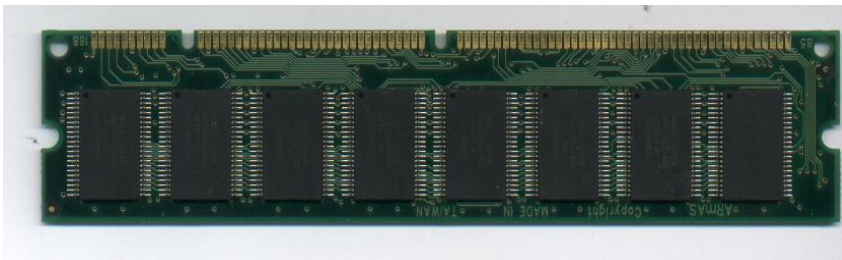
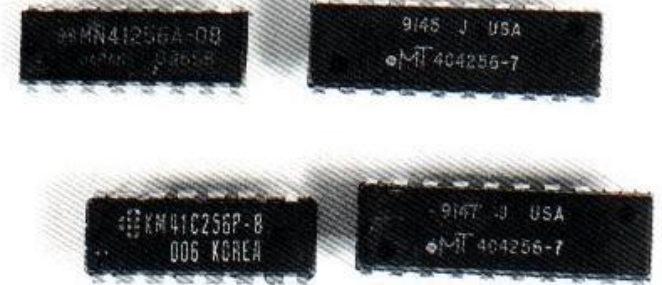
## ■ Storage-device Hierarchy:





# 記憶體 (Memory) (1)

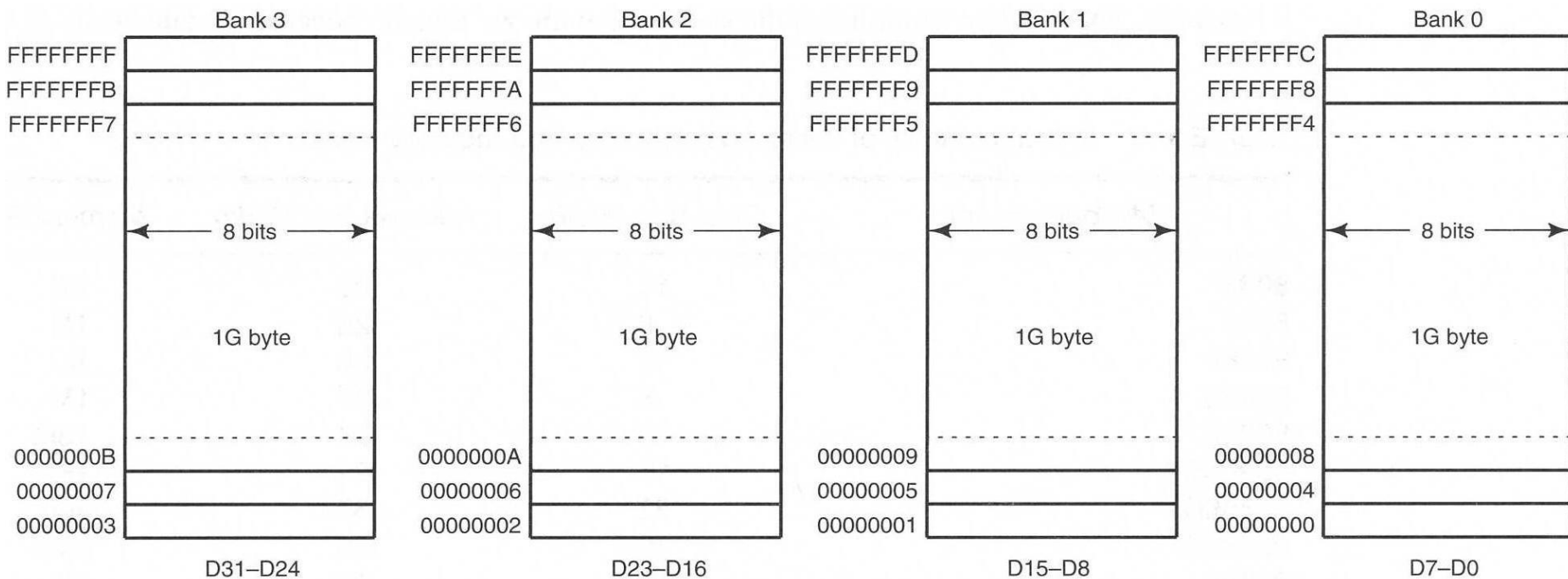
- (1) ROM (Read only memory).
  - PROM, EPROM, EEPROM.
- (2) RAM (Random access memory).
  - (a) DRAM (capacitor, refresh circuit).
    - (1) SDRAM (Synchronous DRAM).
    - (2) DDR (Double Data Rate)：在一個時間週期，同時處理兩筆資料，相當於兩條SDRAM的輸出量，所以執行速度更快。
    - (3) RDRAM (Rambus DRAM).
  - (b) SRAM (flip-flops).





# 記憶體 (Memory) (2)

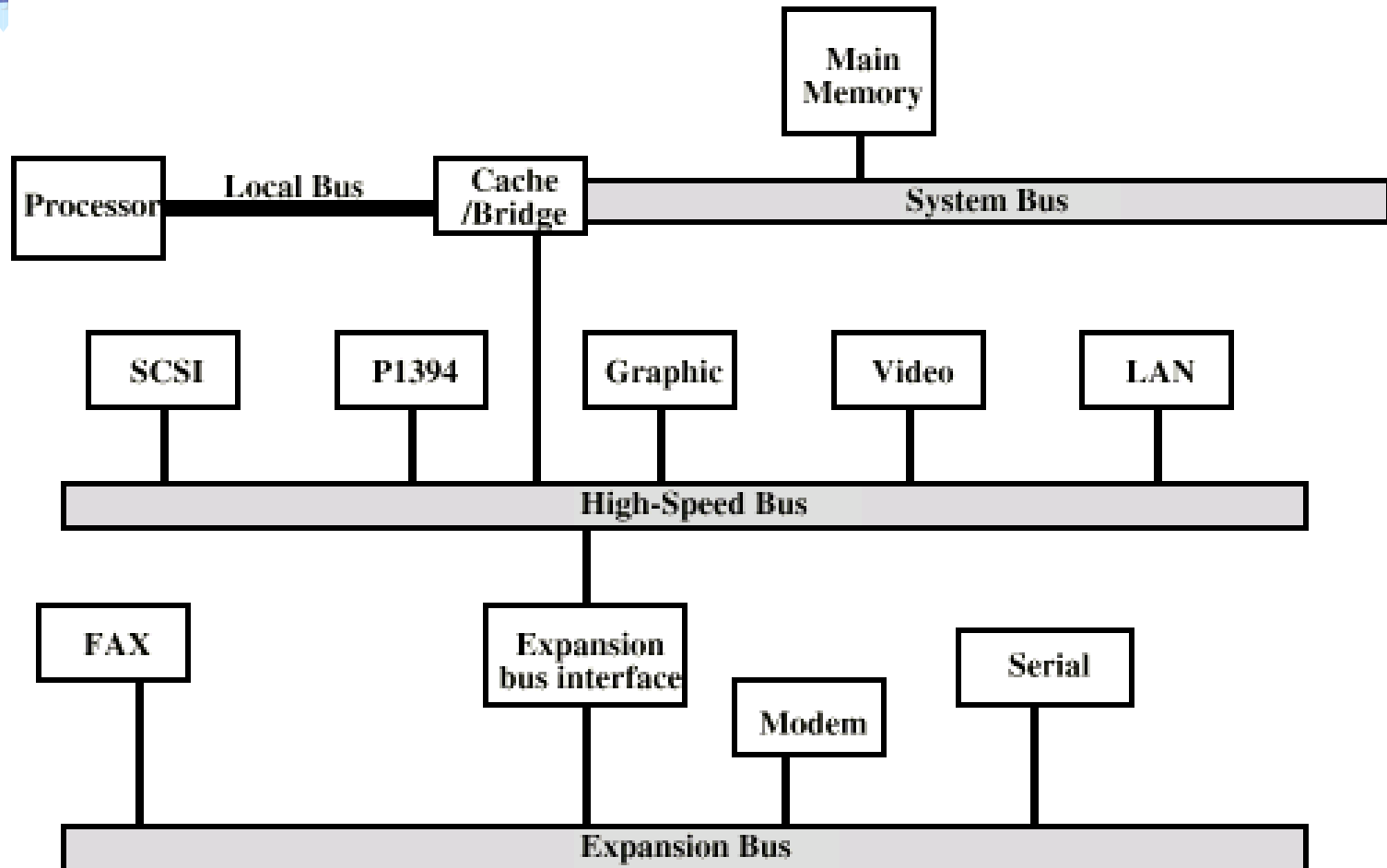
- The physical memory systems of 80386DX through 80486 SX/DX (32-bit data bus) (8-bit = 1-byte).



80386DX microprocessor  
80486SX microprocessor  
80486DX microprocessor

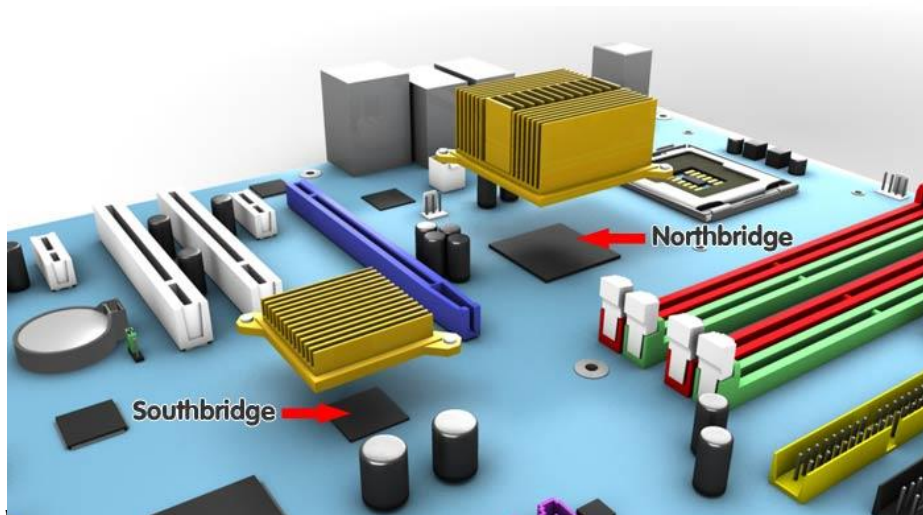
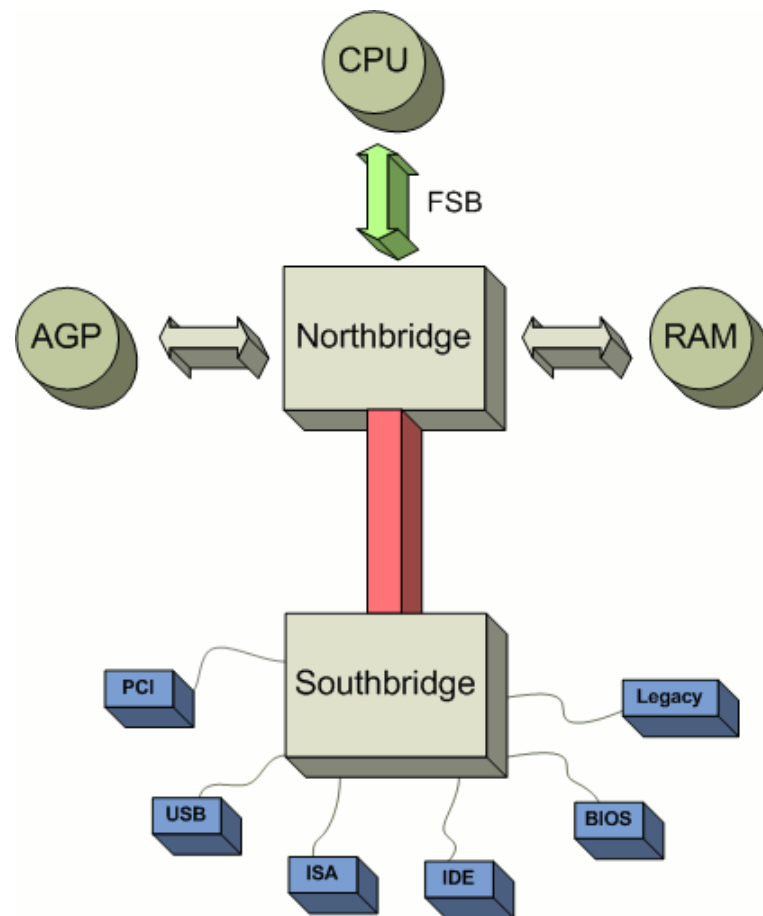
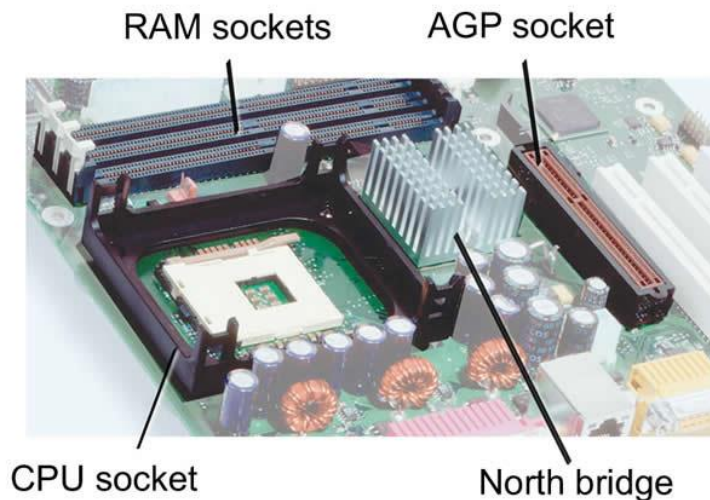


# High Performance Bus Architecture (with Cache)





# North & South Bridges (Chips)

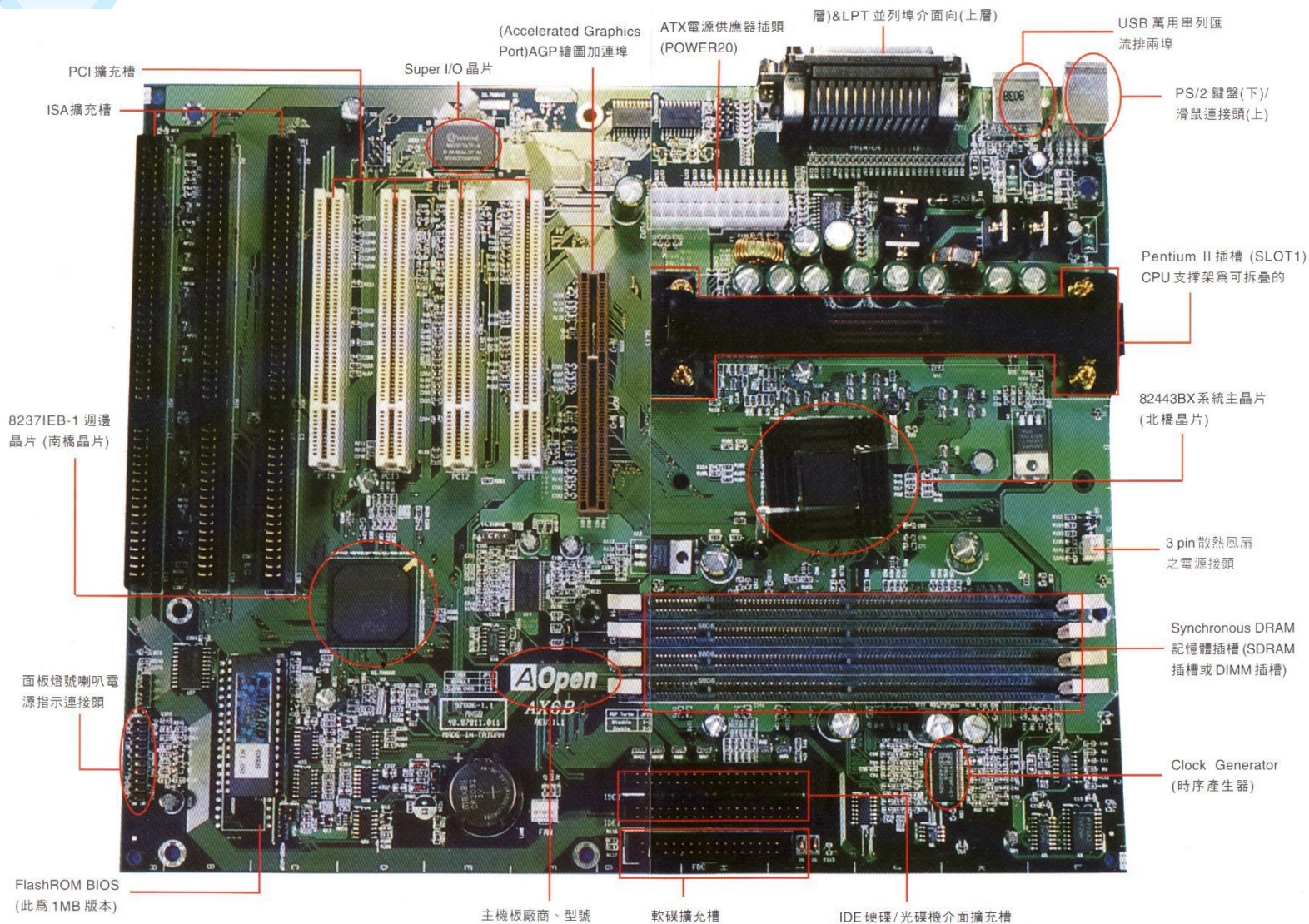






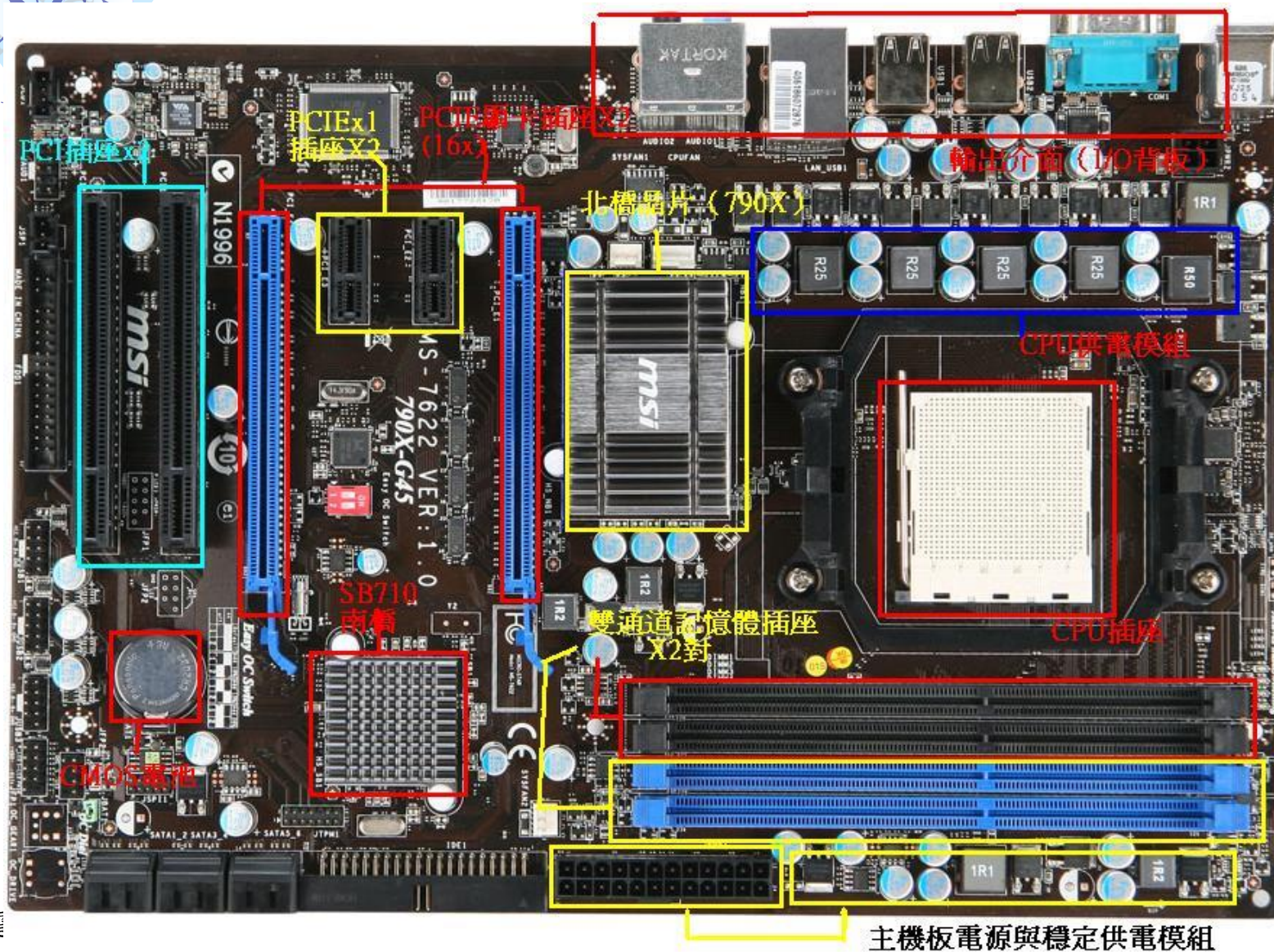
# Main board (2)

## Slot 1 (AX6B)





# Main Board (3)





# Number Systems (1)

- Real = Integer part + Fractional part (實數 = 整數+小數).

- (1) Decimal Number (十進制數值):

Symbols: 0,1,2,3,4,5,6,7,8,9.

Example: the real number  $+24.13 = (+24.13)_{10}$ .

|         |               |   |              |   |                |   |                 |              |
|---------|---------------|---|--------------|---|----------------|---|-----------------|--------------|
|         | $10^1$        |   | $10^0$       |   | $10^{-1}$      |   | $10^{-2}$       | Place values |
|         | 2             |   | 4            | • | 1              |   | 3               | Number       |
| $R = +$ | $2 \times 10$ | + | $4 \times 1$ | + | $1 \times 0.1$ | + | $3 \times 0.01$ | Values       |

- (2) Binary Number (二進制數值):

Symbols: 0,1.

Example: the real number  $(+101.11)_2 = (+5.75)_{10}$ .

|       |                |   |                |   |                |   |                   |   |                   |              |
|-------|----------------|---|----------------|---|----------------|---|-------------------|---|-------------------|--------------|
|       | $2^2$          |   | $2^1$          |   | $2^0$          |   | $2^{-1}$          |   | $2^{-2}$          | Place values |
|       | 1              |   | 0              |   | 1              | • | 1                 |   | 1                 | Number       |
| $R =$ | $1 \times 2^2$ | + | $0 \times 2^1$ | + | $1 \times 2^0$ | + | $1 \times 2^{-1}$ | + | $1 \times 2^{-2}$ | Values       |



# Number Systems (2)

## ■ (3) Octal Number (八進制數值):

Symbols: 0,1,2,3,4,5,6,7.

Example: the real number  $(+1256)_8 = (+686)_{10}$ .

|     |                |   |                |   |                |   |                |              |
|-----|----------------|---|----------------|---|----------------|---|----------------|--------------|
|     | $8^3$          |   | $8^2$          |   | $8^1$          |   | $8^0$          | Place values |
|     | 1              |   | 2              |   | 5              |   | 6              | Number       |
| N = | $1 \times 8^3$ | + | $2 \times 8^2$ | + | $5 \times 8^1$ | + | $6 \times 8^0$ | Values       |

## ■ (4) Hexadecimal Number (十六進制數值):

Symbols: 0,1,2,3,4,5,6,7,8,9,A,B,C,D,E,F.

Example: the real number  $(+2AE)_{16} = (+686)_{10}$ .

|     |                 |   |                  |   |                  |              |
|-----|-----------------|---|------------------|---|------------------|--------------|
|     | $16^2$          |   | $16^1$           |   | $16^0$           | Place values |
|     | 2               |   | A                |   | E                | Number       |
| N = | $2 \times 16^2$ | + | $10 \times 16^1$ | + | $14 \times 16^0$ | Values       |



# Number Systems (3)

- 4 number systems:

Summary of the four positional number systems

| <i>System</i> | <i>Base</i> | <i>Symbols</i>                                 | <i>Examples</i> |
|---------------|-------------|------------------------------------------------|-----------------|
| Decimal       | 10          | 0, 1, 2, 3, 4, 5, 6, 7, 8, 9                   | 2345.56         |
| Binary        | 2           | 0, 1                                           | $(1001.11)_2$   |
| Octal         | 8           | 0, 1, 2, 3, 4, 5, 6, 7                         | $(156.23)_8$    |
| Hexadecimal   | 16          | 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F | $(A2C.A1)_{16}$ |



# ASCII Code (1)

- ASCII (American Standard Code for Information Interchange) data represent alphanumeric characters in a computer system.
- Original, ASCII code is a 7-bit code.
- An extended ASCII code is a 8-bit code.
  - The extended ASCII characters store some foreign letters & punctuation, Greek characters, mathematical characters, and so forth.



# ASCII Code (2)

## ■ Part 1:

| ASCII 碼 |      | 字元 | 控制字元 | 意義              | ASCII 碼 |      | 字元 | 控制字元 | 意義     |
|---------|------|----|------|-----------------|---------|------|----|------|--------|
| 十進位     | 十六進位 |    |      |                 | 十進位     | 十六進位 |    |      |        |
| 000     | 00   |    | NULL | 空字元             | 016     | 10   | ▶  | DLE  |        |
| 001     | 01   | ☺  | SOH  |                 | 017     | 11   | ◀  | DC1  |        |
| 002     | 02   | ☺  | STX  |                 | 018     | 12   | ↕  | DC2  |        |
| 003     | 03   | ♥  | ETX  |                 | 019     | 13   | !! | DC3  |        |
| 004     | 04   | ♦  | EOT  |                 | 020     | 14   | ¶  | DC4  |        |
| 005     | 05   | ♣  | ENQ  |                 | 021     | 15   | §  | NAK  |        |
| 006     | 06   | ♠  | ACK  |                 | 022     | 16   | —  | SYN  |        |
| 007     | 07   | •  | BELL | 鈴聲              | 023     | 17   | ↓  | ETB  |        |
| 008     | 08   | ◻  | BS   | 倒退鍵             | 024     | 18   | ↑  | CAN  |        |
| 009     | 09   |    | HT   | 定位鍵             | 025     | 19   | ↓  | EM   |        |
| 010     | 0A   |    | LF   | line feed       | 026     | 1A   | →  | SUB  | 檔案結束   |
| 011     | 0B   | ♂  | VT   | home            | 027     | 1B   | ←  | ESC  | escape |
| 012     | 0C   | ♀  | FF   | form feed       | 028     | 1C   | L  | FS   | 向右游標   |
| 013     | 0D   |    | CR   | carriage return | 029     | 1D   | ↔  | GS   | 向左游標   |
| 014     | 0E   | ♪  | SO   |                 | 030     | 1E   | ▲  | RS   | 向上游標   |
| 015     | 0F   | ☼  | SI   |                 | 031     | 1F   | ▼  | US   | 向下游標   |



# ASCII Code (3)

## ■ Part 2:

| ASCII 碼 |      | 字元 | ASCII 碼 |      | 字元 | ASCII 碼 |      | 字元 | ASCII 碼 |      | 字元 |
|---------|------|----|---------|------|----|---------|------|----|---------|------|----|
| 十進位     | 十六進位 |    | 十進位     | 十六進位 |    | 十進位     | 十六進位 |    | 十進位     | 十六進位 |    |
| 032     | 20   |    | 056     | 38   | 8  | 080     | 50   | P  | 104     | 68   | h  |
| 033     | 21   | !  | 057     | 39   | 9  | 081     | 51   | Q  | 105     | 69   | i  |
| 034     | 22   | "  | 058     | 3A   | :  | 082     | 52   | R  | 106     | 6A   | j  |
| 035     | 23   | #  | 059     | 3B   | ;  | 083     | 53   | S  | 107     | 6B   | k  |
| 036     | 24   | \$ | 060     | 3C   | <  | 084     | 54   | T  | 108     | 6C   | l  |
| 037     | 25   | %  | 061     | 3D   | =  | 085     | 55   | U  | 109     | 6D   | m  |
| 038     | 26   | &  | 062     | 3E   | >  | 086     | 56   | V  | 110     | 6E   | n  |
| 039     | 27   | '  | 063     | 3F   | ?  | 087     | 57   | W  | 111     | 6F   | o  |
| 040     | 28   | (  | 064     | 40   | @  | 088     | 58   | X  | 112     | 70   | p  |
| 041     | 29   | )  | 065     | 41   | A  | 089     | 59   | Y  | 113     | 71   | q  |
| 042     | 2A   | *  | 066     | 42   | B  | 090     | 5A   | Z  | 114     | 72   | r  |
| 043     | 2B   | +  | 067     | 43   | C  | 091     | 5B   | [  | 115     | 73   | s  |
| 044     | 2C   | ,  | 068     | 44   | D  | 092     | 5C   | \  | 116     | 74   | t  |
| 045     | 2D   | -  | 069     | 45   | E  | 093     | 5D   | ]  | 117     | 75   | u  |
| 046     | 2E   | .  | 070     | 46   | F  | 094     | 5E   | ^  | 118     | 76   | v  |
| 047     | 2F   | /  | 071     | 47   | G  | 095     | 5F   | _  | 119     | 77   | w  |
| 048     | 30   | 0  | 072     | 48   | H  | 096     | 60   | `  | 120     | 78   | x  |
| 049     | 31   | 1  | 073     | 49   | I  | 097     | 61   | a  | 121     | 79   | y  |
| 050     | 32   | 2  | 074     | 4A   | J  | 098     | 62   | b  | 122     | 7A   | z  |
| 051     | 33   | 3  | 075     | 4B   | K  | 099     | 63   | c  | 123     | 7B   | {  |
| 052     | 34   | 4  | 076     | 4C   | L  | 100     | 64   | d  | 124     | 7C   |    |
| 053     | 35   | 5  | 077     | 4D   | M  | 101     | 65   | e  | 125     | 7D   | }  |
| 054     | 36   | 6  | 078     | 4E   | N  | 102     | 66   | f  | 126     | 7E   | ~  |
| 055     | 37   | 7  | 079     | 4F   | O  | 103     | 67   | g  | 127     | 7F   | ␣  |



# ASCII Code (4)

## ■ Part 3:

| ASCII 碼 |      |  | 字元 | ASCII 碼 |      |  | 字元 | ASCII 碼 |      |  | 字元 | ASCII 碼 |      |  | 字元 |
|---------|------|--|----|---------|------|--|----|---------|------|--|----|---------|------|--|----|
| 十進位     | 十六進位 |  |    | 十進位     | 十六進位 |  |    | 十進位     | 十六進位 |  |    | 十進位     | 十六進位 |  |    |
| 128     | 80   |  | Ç  | 160     | A0   |  | á  | 192     | C0   |  | Ł  | 224     | E0   |  | α  |
| 129     | 81   |  | ü  | 161     | A1   |  | í  | 193     | C1   |  | ł  | 225     | E1   |  | β  |
| 130     | 82   |  | é  | 162     | A2   |  | ó  | 194     | C2   |  | ṽ  | 226     | E2   |  | γ  |
| 131     | 83   |  | â  | 163     | A3   |  | ú  | 195     | C3   |  | ṽ  | 227     | E3   |  | π  |
| 132     | 84   |  | ä  | 164     | A4   |  | ñ  | 196     | C4   |  | —  | 228     | E4   |  | Σ  |
| 133     | 85   |  | à  | 165     | A5   |  | Ñ  | 197     | C5   |  | †  | 229     | E5   |  | σ  |
| 134     | 86   |  | â  | 166     | A6   |  | ª  | 198     | C6   |  | ‡  | 230     | E6   |  | μ  |
| 135     | 87   |  | ç  | 167     | A7   |  | º  | 199     | C7   |  | ‡  | 231     | E7   |  | Τ  |
| 136     | 88   |  | ê  | 168     | A8   |  | ¿  | 200     | C8   |  | Ł  | 232     | E8   |  | φ  |
| 137     | 89   |  | ë  | 169     | A9   |  | ƒ  | 201     | C9   |  | Ŕ  | 233     | E9   |  | θ  |
| 138     | 8A   |  | è  | 170     | AA   |  | ¬  | 202     | CA   |  | Ŕ  | 234     | EA   |  | Ω  |
| 139     | 8B   |  | ï  | 171     | AB   |  | ½  | 203     | CB   |  | Ť  | 235     | EB   |  | δ  |
| 140     | 8C   |  | î  | 172     | AC   |  | ¼  | 204     | CC   |  | ‡  | 236     | EC   |  | ∞  |
| 141     | 8D   |  | ì  | 173     | AD   |  | ı  | 205     | CD   |  | =  | 237     | ED   |  | φ  |
| 142     | 8E   |  | Ä  | 174     | AE   |  | «  | 206     | CE   |  | ‡  | 238     | EE   |  | €  |
| 143     | 8F   |  | Å  | 175     | AF   |  | »  | 207     | CF   |  | ⊥  | 239     | EF   |  | ∩  |
| 144     | 90   |  | É  | 176     | B0   |  | ⋮  | 208     | D0   |  | ⊥  | 240     | F0   |  | ≡  |
| 145     | 91   |  | æ  | 177     | B1   |  | ⋮  | 209     | D1   |  | Ŧ  | 241     | F1   |  | ±  |
| 146     | 92   |  | Æ  | 178     | B2   |  | ⋮  | 210     | D2   |  | Ŧ  | 242     | F2   |  | ≥  |
| 147     | 93   |  | ô  | 179     | B3   |  |    | 211     | D3   |  | Ŧ  | 243     | F3   |  | ≤  |
| 148     | 94   |  | ö  | 180     | B4   |  | †  | 212     | D4   |  | Ł  | 244     | F4   |  | ∫  |
| 149     | 95   |  | ò  | 181     | B5   |  | ‡  | 213     | D5   |  | Ŕ  | 245     | F5   |  | ∫  |
| 150     | 96   |  | û  | 182     | B6   |  | ‡  | 214     | D6   |  | Ŕ  | 246     | F6   |  | ÷  |
| 151     | 97   |  | ù  | 183     | B7   |  | π  | 215     | D7   |  | ‡  | 247     | F7   |  | ≈  |
| 152     | 98   |  | ÿ  | 184     | B8   |  | ƒ  | 216     | D8   |  | ‡  | 248     | F8   |  | °  |



# ASCII Code (5)

## ■ Part 4:

|     |    |   |     |    |   |     |    |   |     |    |   |
|-----|----|---|-----|----|---|-----|----|---|-----|----|---|
| 153 | 99 | Ö | 185 | B9 | Ɔ | 217 | D9 | Ɔ | 249 | F9 | • |
| 154 | 9A | Ü | 186 | BA |   | 218 | DA | Ɔ | 250 | FA | · |
| 155 | 9B | Ɔ | 187 | BB | Ɔ | 219 | DB | ■ | 251 | FB | √ |
| 156 | 9C | Ɔ | 188 | BC | Ɔ | 220 | DC | ■ | 252 | FC | n |
| 157 | 9D | ¥ | 189 | BD | Ɔ | 221 | DD | ■ | 253 | FD | z |
| 158 | 9E | Ɔ | 190 | BE | Ɔ | 222 | DE | ■ | 254 | FE | ■ |
| 159 | 9F | f | 191 | BF | Ɔ | 223 | DF | ■ | 255 | FF |   |

- ASCII 字元表 (<http://home.educities.edu.tw/wanker742126/asm/ap04.html>).